

*MA.8.NSO.1.2***Benchmark**

MA.8.NSO.1.2 Plot, order and compare rational and irrational numbers, represented in various forms.

Benchmark Clarifications:

Clarification 1: Within this benchmark, it is not the expectation to work with the number e .

Clarification 2: Within this benchmark, the expectation is to plot, order and compare square roots and cube roots.

Clarification 3: Within this benchmark, the expectation is to use symbols ($<$, $>$ or $=$).

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.1.2
- MA.8.AR.2.3
- MA.8.GR.1.1, MA.8.GR.1.2

Terms from the K-12 Glossary

- Irrational numbers
- Rational numbers

Vertical Alignment**Previous Benchmarks**

- MA.6.NSO.1.2

Next Benchmarks

- MA.912.NSO.1.4
- MA.912.GR.3.1

Purpose and Instructional Strategies

In previous courses, students were expected to plot, order and compare rational numbers. Students also expressed rational numbers with terminating and repeating decimals. In grade 7 accelerated, students define irrational numbers, recognizing and expressing them in various forms, and students compare rational numbers to irrational numbers. In Algebra 1, students will perform operations with radicals. In Geometry, students will extend their understanding of radical approximations to weighted averages on a number line.

- Students should have the opportunity to draw number lines with appropriate scales to plot the numbers to provide an understanding of where the numbers are in relation to numbers that are greater than and less than the number to be plotted.
- Students locate and compare rational and irrational numbers on the number line. Additionally, students understand that the value of a square root or a cube root can be approximated between integers.
 - For example, to find an approximation of $\sqrt{28}$, two methods are described below, each using the nearest perfect squares to the radicand:
 - determine the perfect squares 28 is between, which would be 25 and 36. The square roots of 25 and 36 are 5 and 6, respectively, so we know that $\sqrt{28}$ is between 5 and 6. Since 28 is closer to 25, an estimate of the square root would be closer to 5.

- since 28 is located $\frac{3}{11}$ of the distance from 25 to 36, the $\sqrt{28}$ is approximately located $\frac{3}{11}$ of the distance from $\sqrt{25}$ to $\sqrt{36}$. So, this reasoning gives the approximation $\sqrt{25} + \frac{3}{11}$, which is about 5.27. This method is particularly relevant when students determine weighted averages on a number line in Geometry.
- Students also recognize that every positive number has both a positive and a negative square root. The negative square root of n is written as $-\sqrt{n}$.
- Instruction includes the use of technology, including a calculator.

Common Misconceptions or Errors

- Students may not understand that square and cube roots can be plotted on a number line.

Strategies to Support Tiered Instruction

- Instruction includes providing students with examples of square and cube roots for them to place on a number line and facilitating a conversation on understanding the value of each square and cube root.
- Teacher provides opportunities to co-create number lines with appropriate scales and plot approximate values of cube roots and square roots.
 - For example, provide partially completed examples of non-perfect square roots and non-perfect cube roots by using perfect square roots and perfect cube roots as benchmark quantities.
- Teacher provides support in recognizing that every positive number has both a positive and negative square root.
 - For example, show examples of how multiplying two negative numbers gives a positive number, and how the square root of a number can be both positive and negative.
- Teacher assists students in writing an inequality to represent written statements.
 - For example:
 - Katelyn has more books than Mary.
 - Cameron has 5 pencils and Barry has 8.
 - Animal Kingdom has at least 100 different species of animals.

Instructional Tasks

Instructional Task 1 (MTR.6.1)

Below are irrational and rational numbers.

$$1.\bar{3} \quad \pi \quad \frac{3}{2} \quad \sqrt{3} \quad 2.356 \quad \sqrt{6} \quad \sqrt{4}$$

Part A. Order the numbers from least to greatest by plotting on a number line.

Part B. Identify which numbers are irrational.

Part C. Write an inequality that compares a rational number and an irrational number from the list.

Instructional Task 2 (MTR.3.1)

Haylie is comparing $\sqrt{8}$ and $\sqrt[3]{9}$.

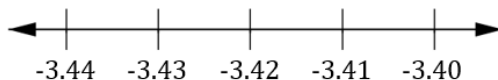
Part A. Describe how Haylie would approximate $\sqrt{8}$ and $\sqrt[3]{9}$.

Part B. Write an inequality for both numbers with the closest rational number.

Instructional Items

Instructional Item 1

Plot $-3.42857\dots$ on the number line below and explain how you determined its location.

*Instructional Item 2*

Using the chart below, compare the irrational and rational numbers shown.

Number	Write < or >	Number
π^2		9
$\sqrt{50}$		7
$\frac{5}{3}$		$\sqrt{8}$
-2π		-6

Instructional Item 3

Plot the following cube roots on a number line $\sqrt[3]{8}$, $\sqrt[3]{10}$ and $\sqrt[3]{27}$.

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*